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Escuela de Ciencias Aplicadas e Ingeniería

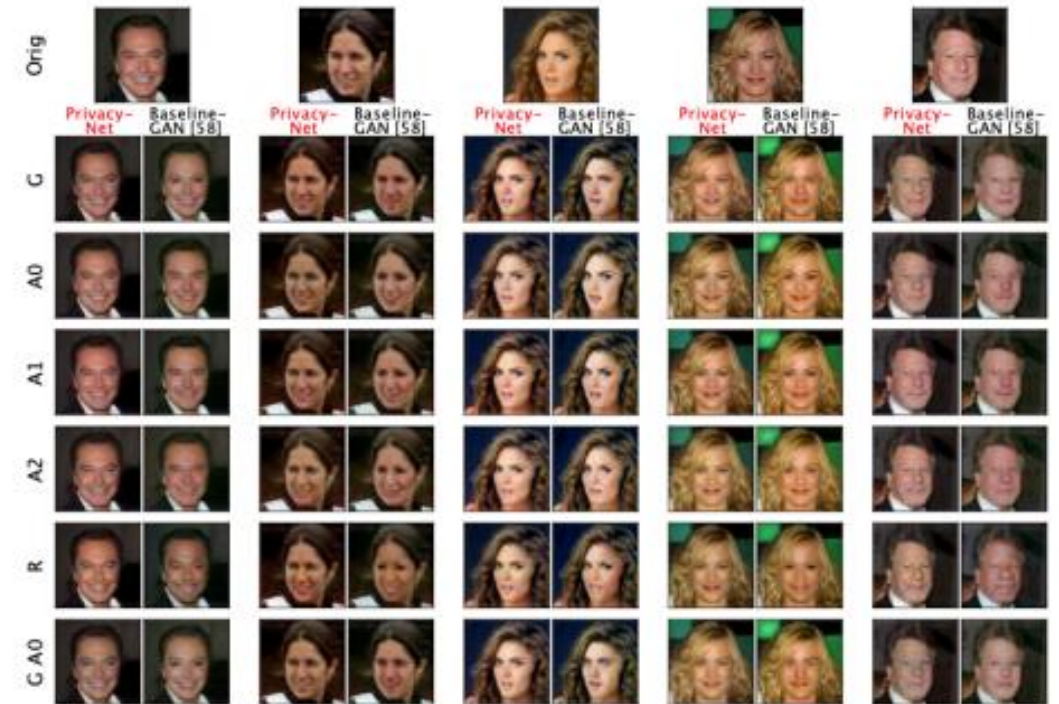
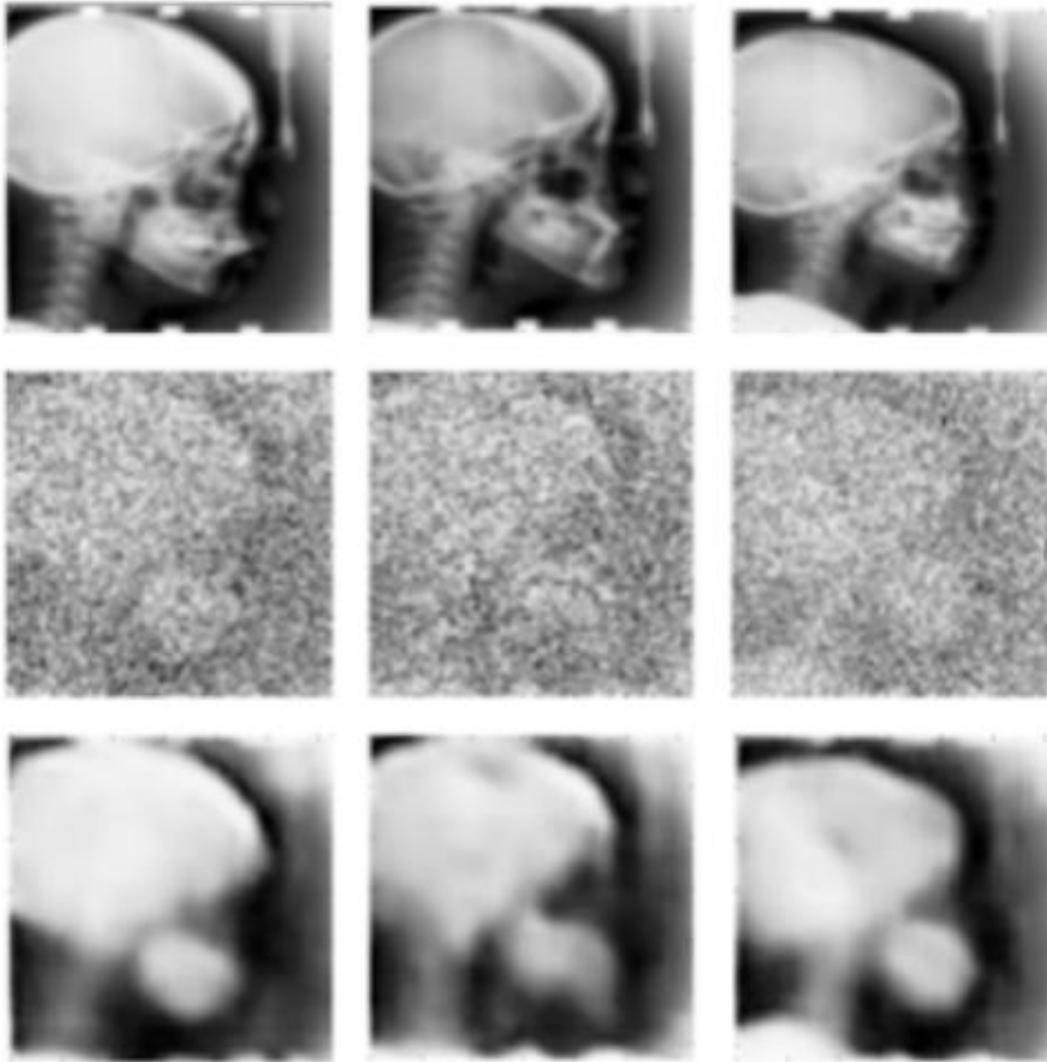
Lecture 09

Deep Learning

Agenda

- **Autoencoders**
- **Convolutional Autoencoders**

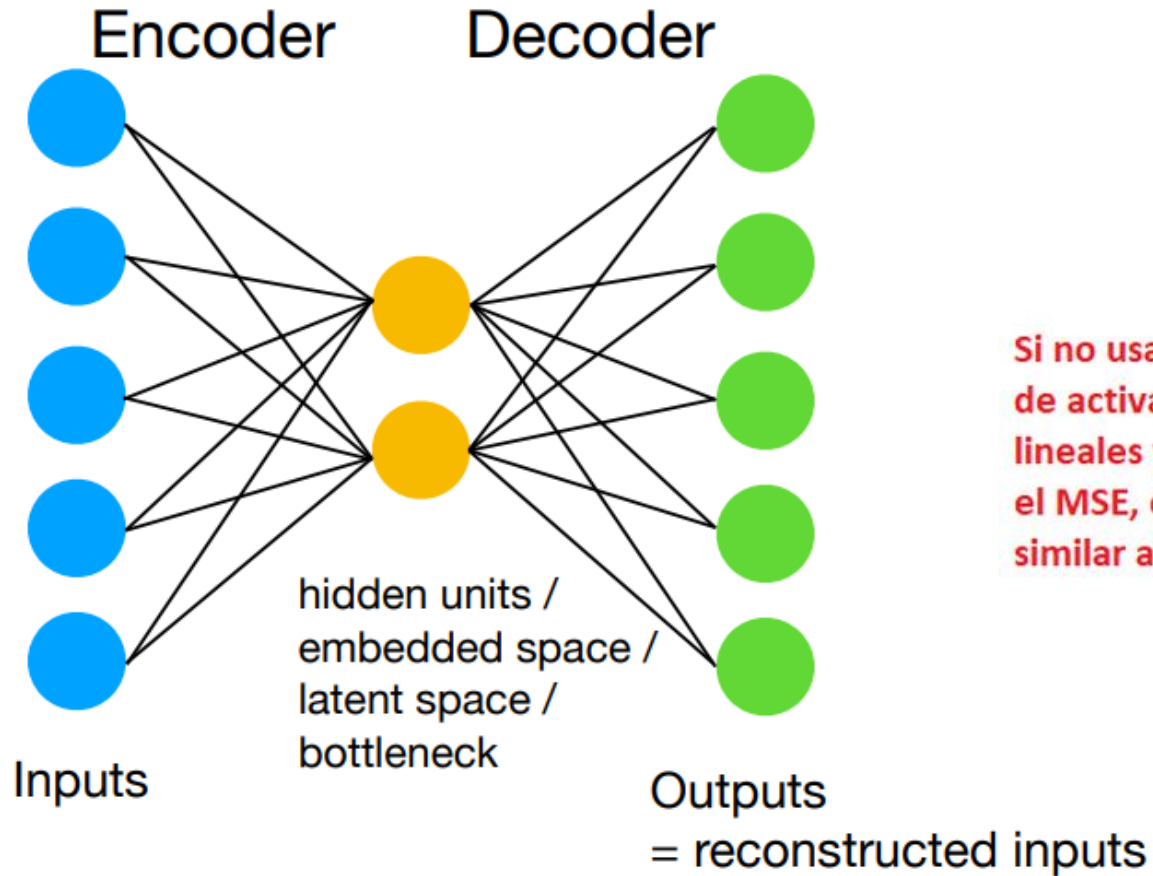
AutoEncoders



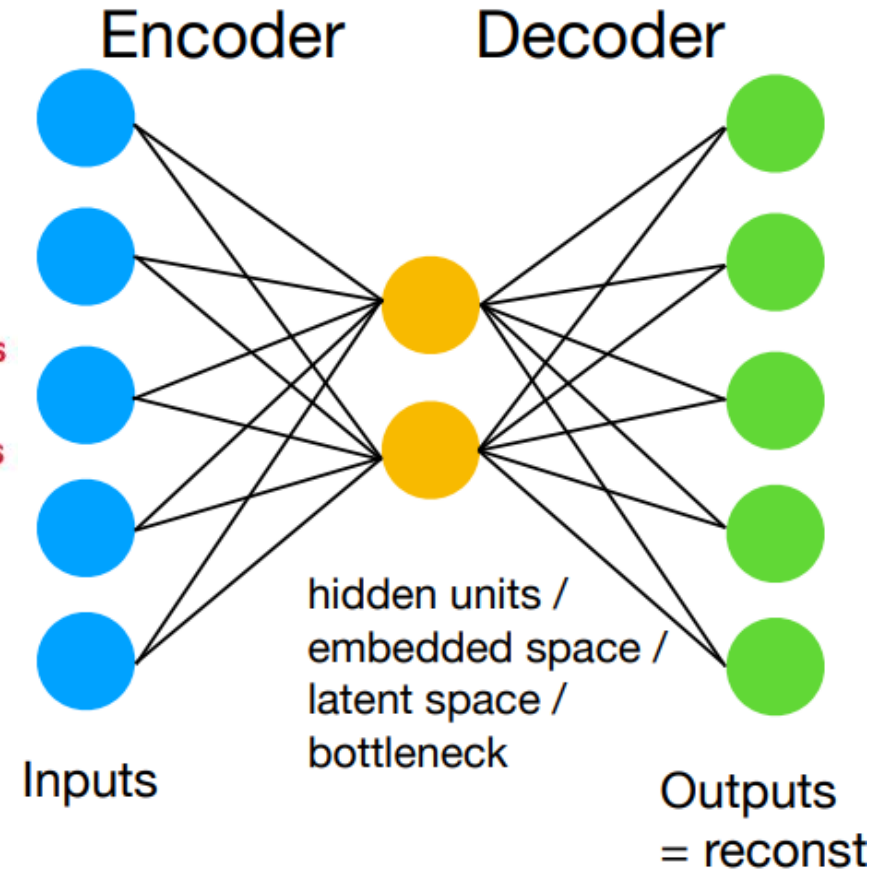
"About half (52%) of U.S. adults said they decided recently not to use a product or service because they were worried about how much personal information would be collected about them."

<https://www.pewresearch.org/fact-tank/2020/04/14/half-of-americans-have-decided-not-to-use-a-product-or-service-because-of-privacy-concerns/>

AutoEncoders completamente conectados

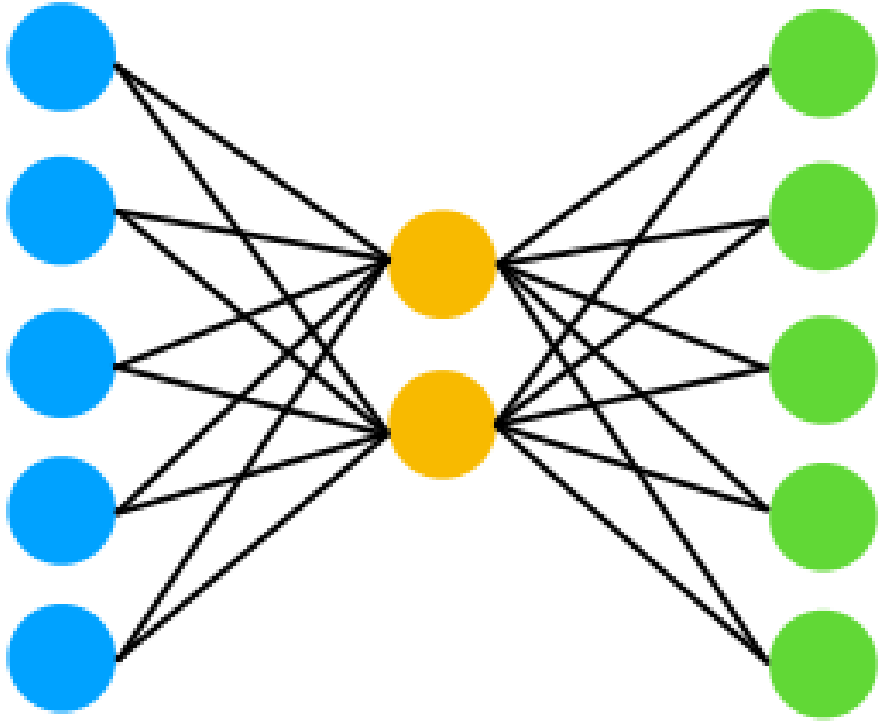


Si no usamos funciones de activación no lineales y minimizamos el MSE, esto es muy similar al PCA



$$\mathcal{L}(\mathbf{x}, \mathbf{x}') = ||\mathbf{x} - \mathbf{x}'||_2^2 = \sum_i (x_i - x'_i)^2$$

AutoEncoders completamente conectados

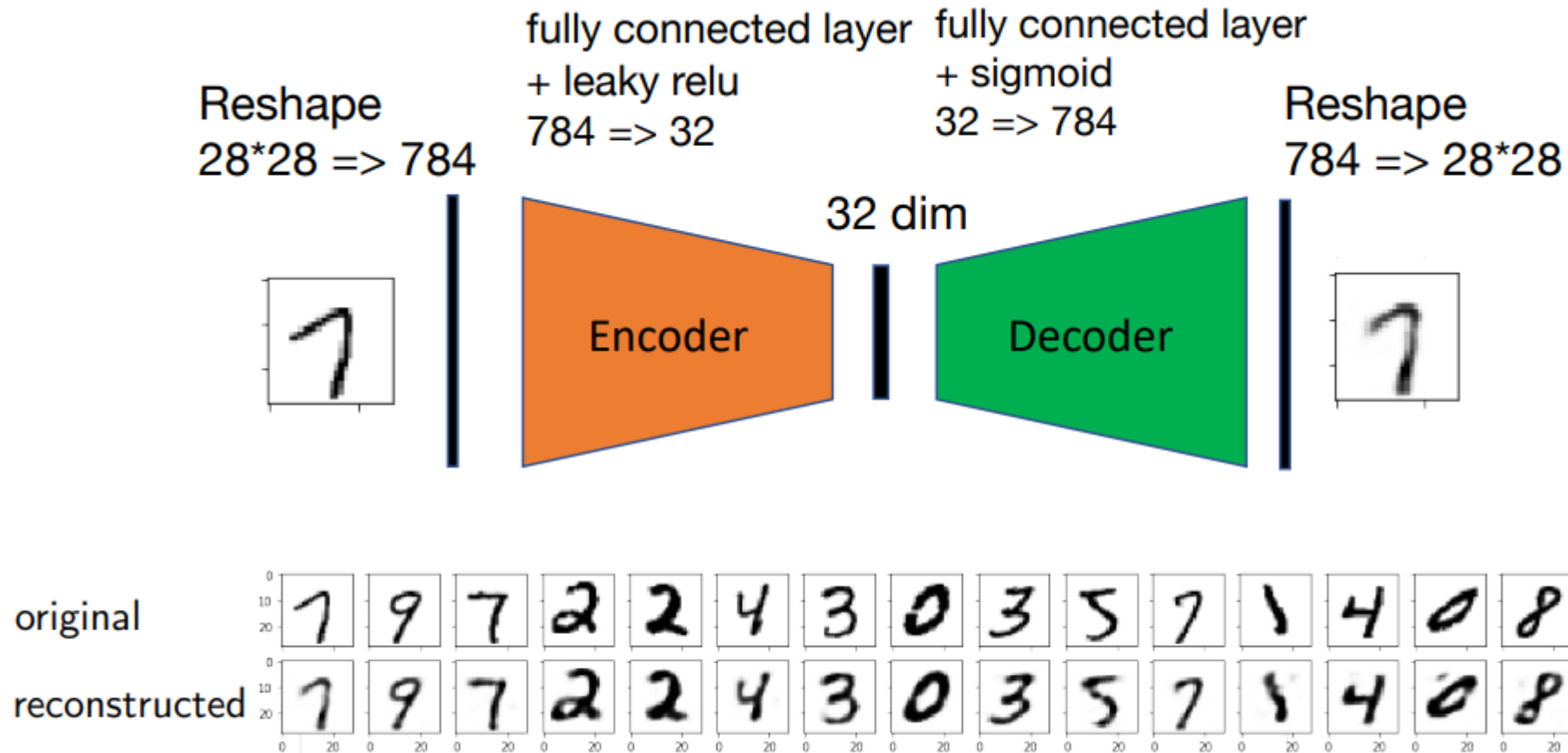


Pregunta:

Si podemos lograr lo mismo con PCA, que es esencialmente un tipo de factorización matricial que es más eficiente que Backprop + SGD, ¿por qué molestarse con los codificadores automáticos?

$$\mathcal{L}(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \mathbf{x}'\|_2^2 = \sum_i (x_i - x'_i)^2$$

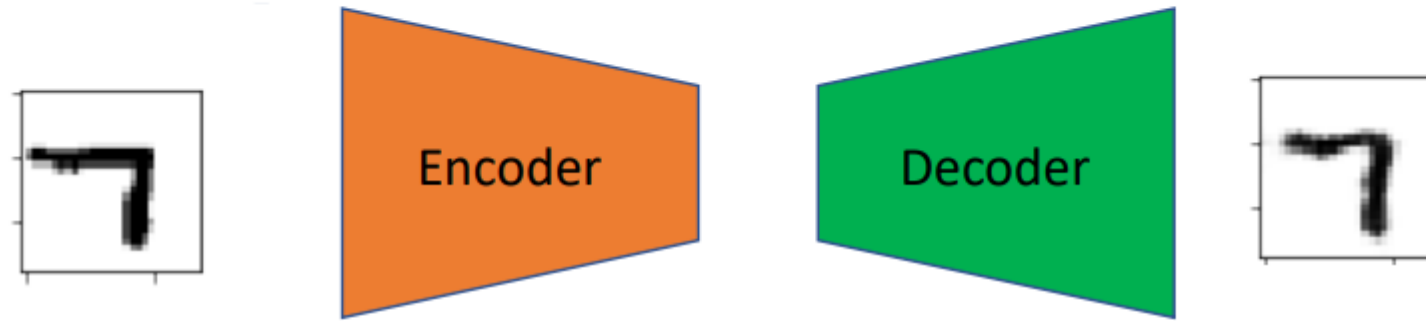
AutoEncoders completamente conectados



AutoEncoders Convolucionales

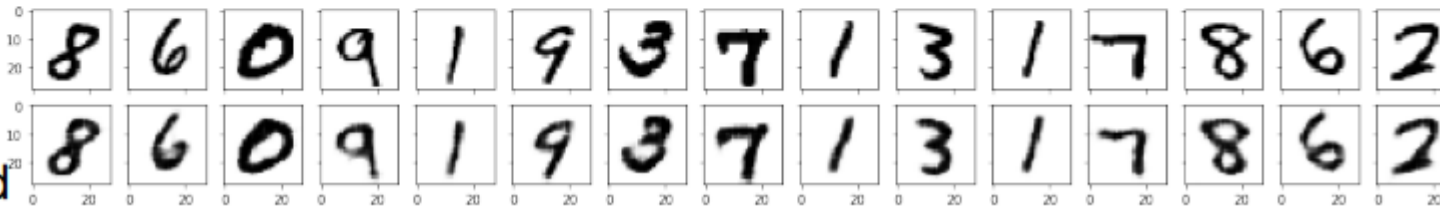
1 or more
convolutional layers

1 or more
"de"convolutional layers

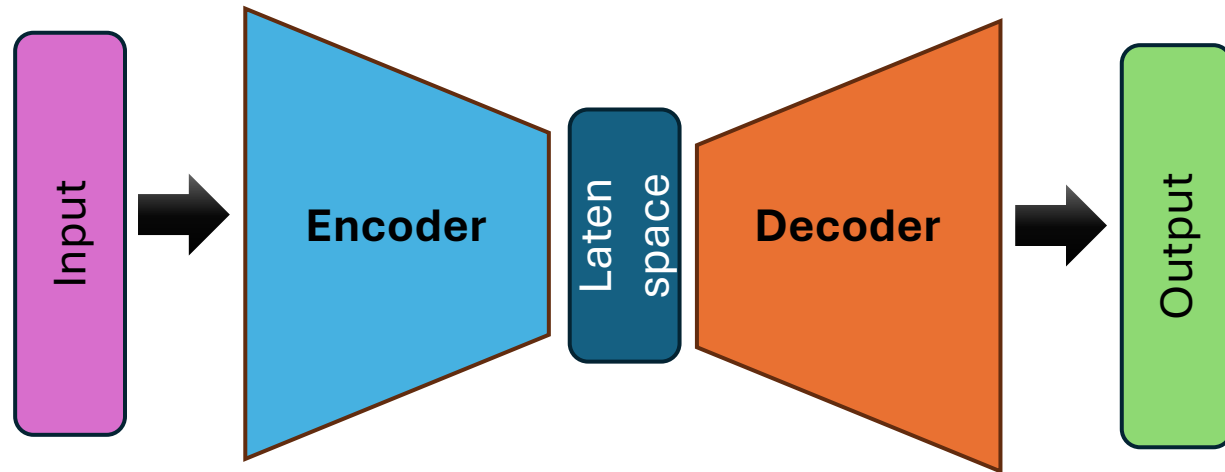


original

reconstructed



AutoEncoders Convolucionales



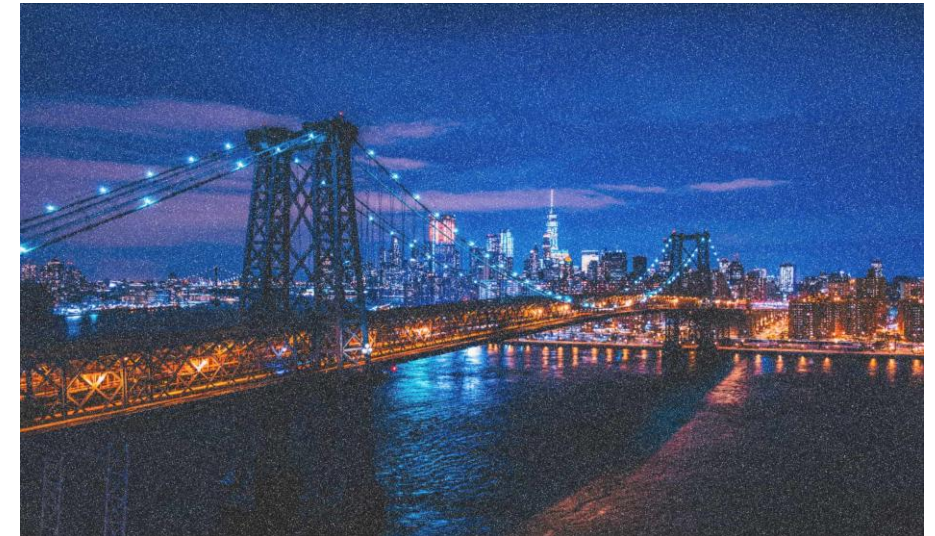
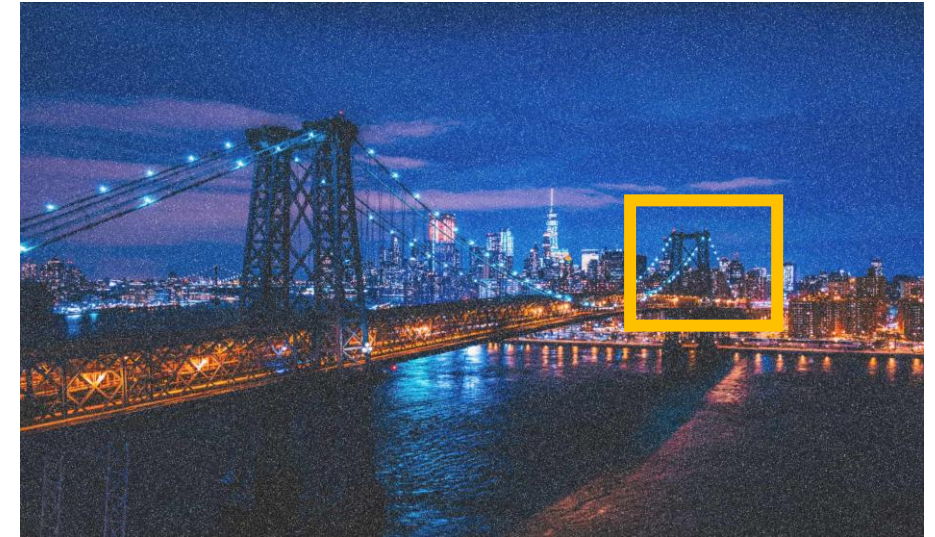
$$x \in \mathcal{R}^{m \times n}$$

$$z \in \mathcal{R}^p$$
$$p \ll m \times n$$

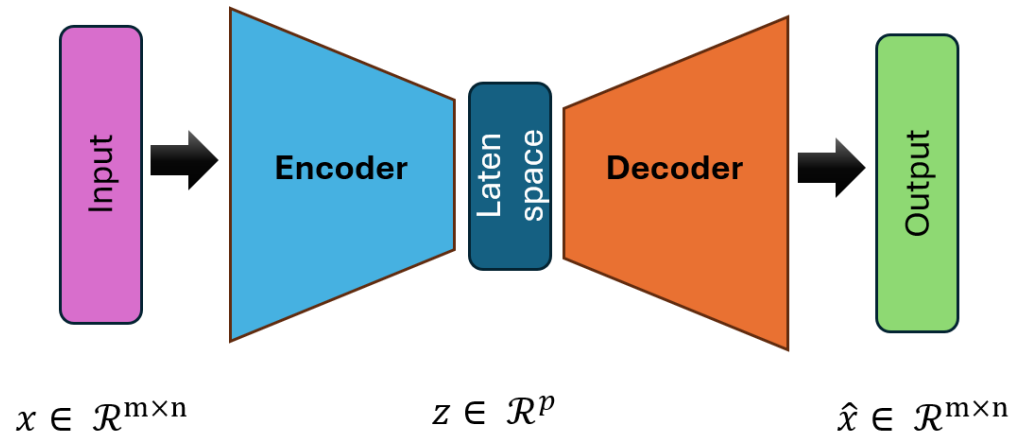
$$\hat{x} \in \mathcal{R}^{m \times n}$$

$$\mathcal{L} = \|x - \hat{x}\|^2$$

!CUIDADO; no deseamos que $x = \hat{x}$



AutoEncoders Convolucionales



$$\mathcal{L} = \|x - \hat{x}\|^2$$

$$z = f_{\theta}(x)$$

$$\hat{x} = g_{\phi}(z)$$

$$\theta = (w_1^e, b_1^e, w_2^e, b_1^e, \dots, w_n^e, b_n^e)$$

$$\phi = (w_1^d, b_1^d, w_2^d, b_1^d, \dots, w_n^d, b_n^d)$$

$$\mathcal{L}_i = \|x_i - \hat{x}_i\|^2$$

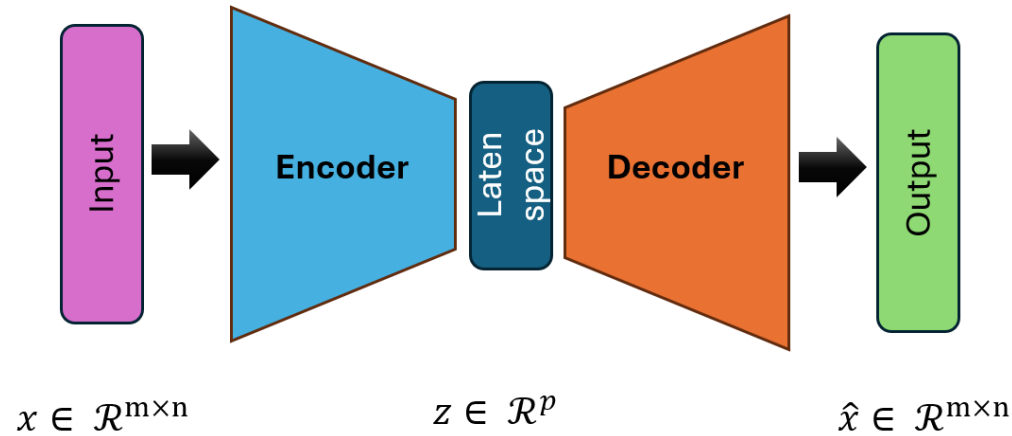
$$J(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}\{x_i, \hat{x}_i\}$$

$$J(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}\{x_i, g_{\phi}(f_{\theta}(x_i))\}$$

$$\theta^*, \phi^* = \arg \min_{\theta, \phi} J(\theta, \phi)$$

$$\theta^*, \phi^* = \arg \min_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \mathcal{L}\{x_i, g_{\phi}(f_{\theta}(x_i))\}$$

AutoEncoders Convolucionales

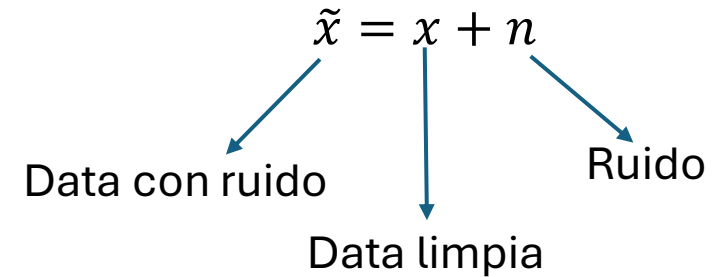


$$\mathcal{L} = \|x - \hat{x}\|^2$$

$$z = f_{\theta}(x) \quad \theta^*, \phi^* = \arg \min_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \mathcal{L}\{x_i, g_{\phi}(f_{\theta}(x_i))\}$$

$$\hat{x} = g_{\phi}(z)$$

!CUIDADO! no deseamos que la salida sea igual a la entrada



$$z = f_{\theta}(\tilde{x}) \Rightarrow \hat{x} = g_{\phi}(z) \Rightarrow \hat{x} = g_{\phi}(f_{\theta}(\tilde{x}))$$

$$\mathcal{L} = \|x - \hat{x}\|^2$$

$$\mathcal{L} = \|x - g_{\phi}(f_{\theta}(\tilde{x}))\|^2$$

$$\mathcal{L} = \|x - (x + n)\|^2$$

$$\mathcal{L} = \|-n\|^2$$

$$\mathcal{L} = \|n\|^2$$

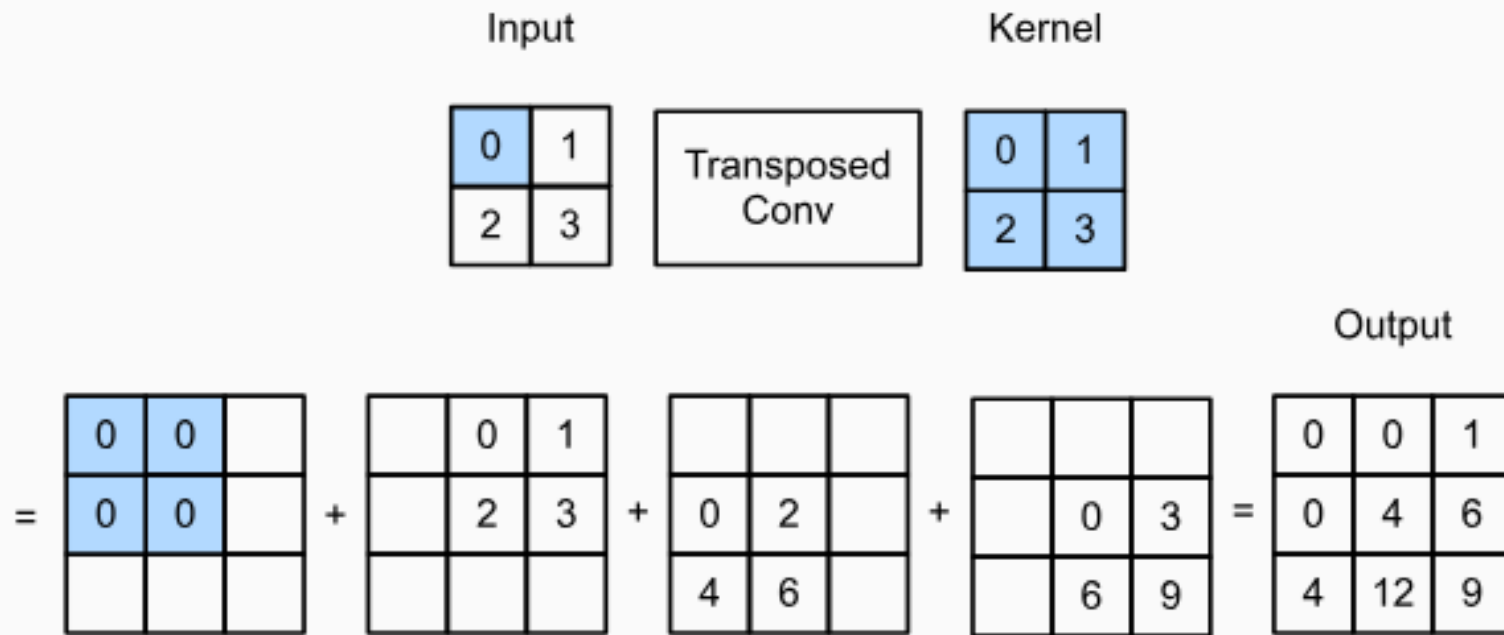
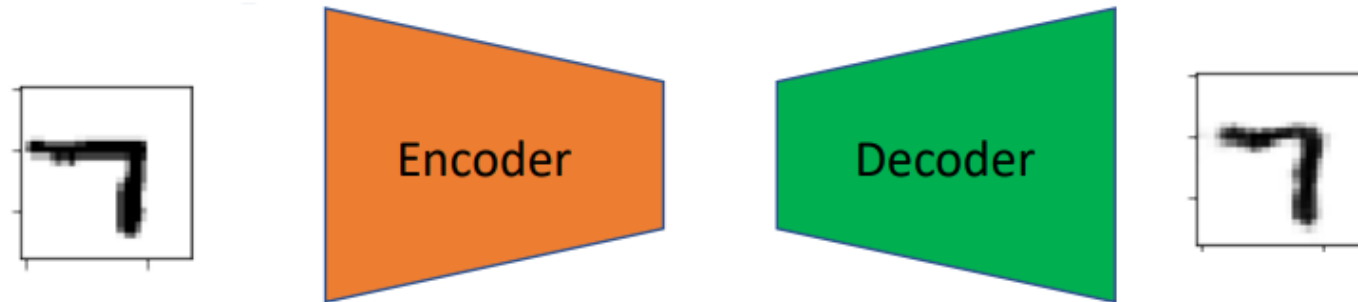
$$\mathcal{L} = \|x - \hat{x}\|^2$$



$$\mathcal{L} = \|0\|^2$$

AutoEncoders Convolucionales

Convolución Transpuesta

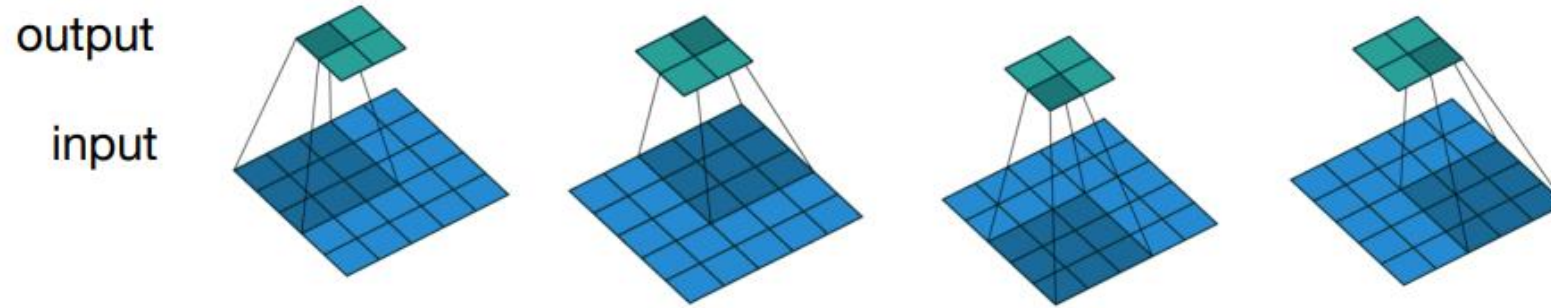


convoluciones transpuestas generan o amplían las dimensiones de los datos, como generar imágenes de mayor resolución a partir de imágenes de menor resolución

$$H_{out} = (H_{in} - 1) \times stride + kernel - 2 \times padding$$

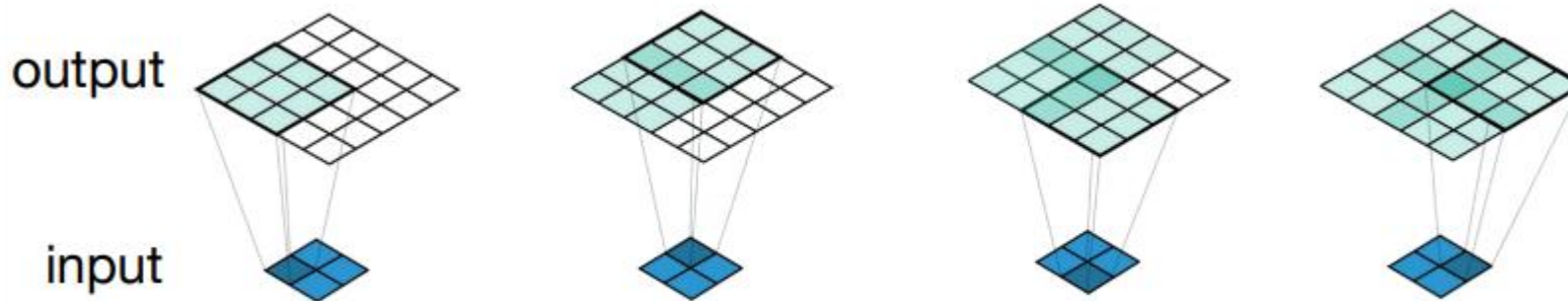
AutoEncoders Convolucionales

Convolución regular



Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." arXiv preprint arXiv:1603.07285 (2016).
<https://arxiv.org/abs/1603.07285>

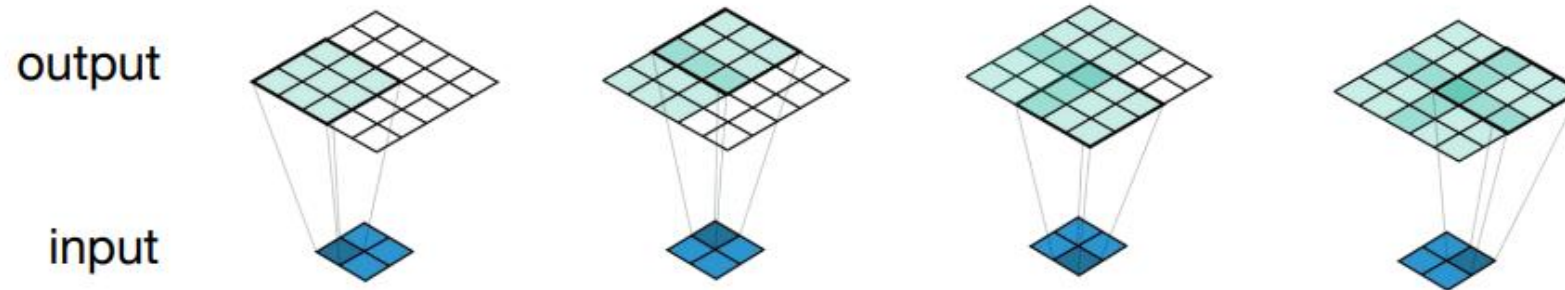
Convolución transpuesta (paso = 2):



A Conv2DTranspose with 3x3 kernel and stride of 2x2 applied to a 2x2 input to give a 5x5 output.
(<https://medium.com/apache-mxnet/transposed-convolutions-explained-with-ms-excel-52d13030c7e8>)

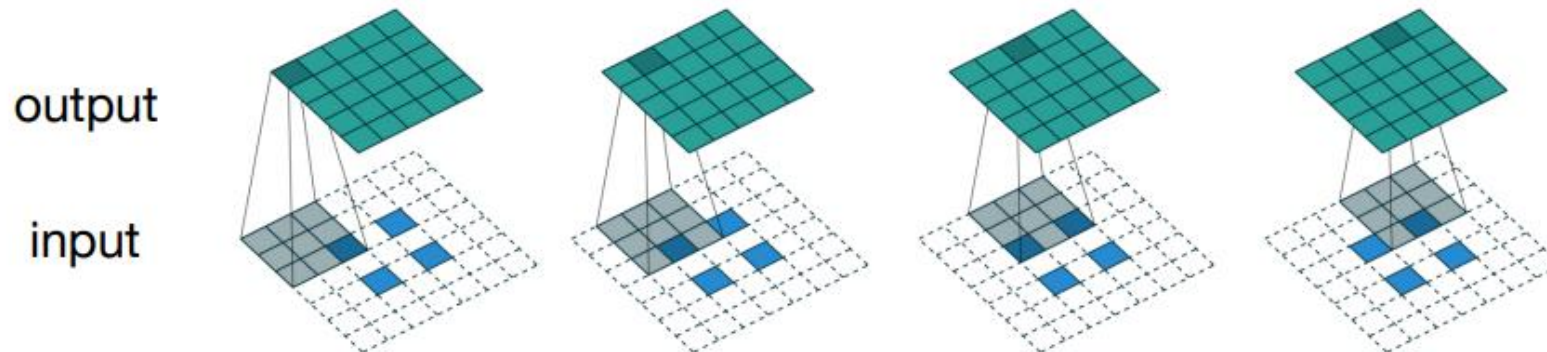
AutoEncoders Convolucionales

Convolución Transpuesta kernel 3 x3, paso 2



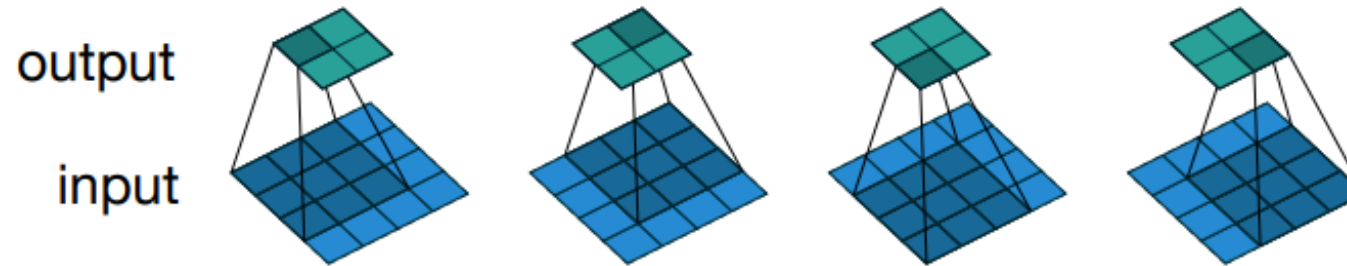
A Conv2DTranspose with 3x3 kernel and stride of 2x2 applied to a 2x2 input to give a 5x5 output.
(<https://medium.com/apache-mxnet/transposed-convolutions-explained-with-ms-excel-52d13030c7e8>)

Convolución Transpuesta (emulada con convolución directa)

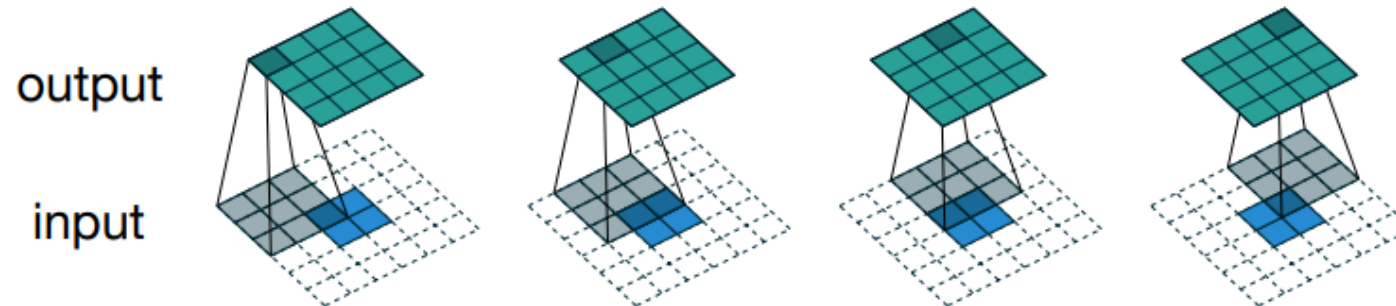


AutoEncoders Convolucionales

Convolución regular: (paso = 1):



Convolución transpuesta (emulada con convolución directa): (paso = 1)

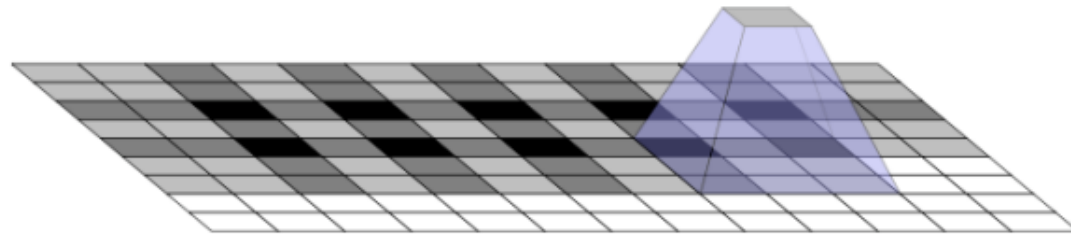


Deconvolution and Checkerboard Artifacts

AUGUSTUS ODENA VINCENT DUMOULIN CHRIS OLAH Oct. 17 Citation:
Google Brain Université de Montréal Google Brain 2016 Odena, et al., 2016

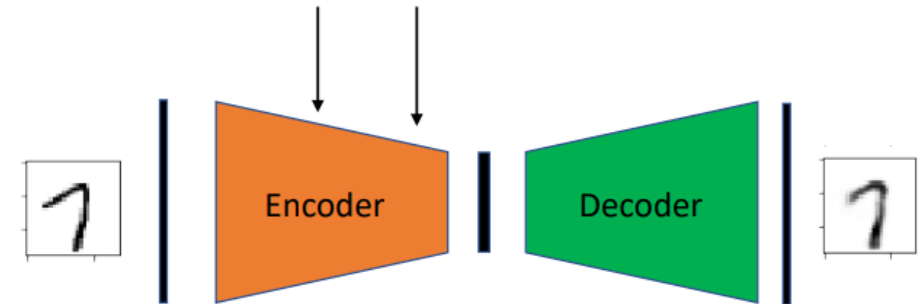
Un buen artículo interactivo que destaca los peligros de la convolución transpuesta.

<https://distill.pub/2016/deconv-checkerboard/>



En resumen, recomienda reemplazar la convolución transpuesta por muestreo ascendente (interpolación) seguido de convolución regular

Agregue capas de dropout para obligar a las redes a aprender características redundantes

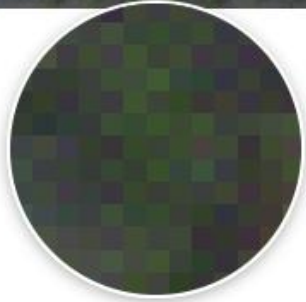
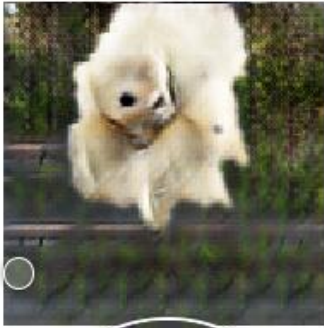


AutoEncoders Convolucionales

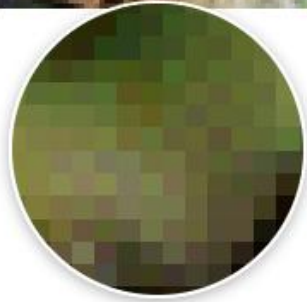
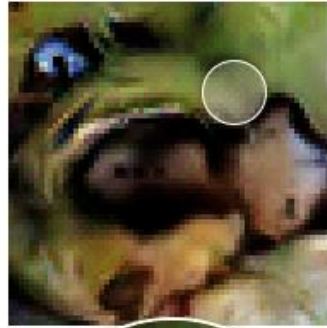
[Link](#)



Radford, et al., 2015 [1]



Salimans et al., 2016 [2]



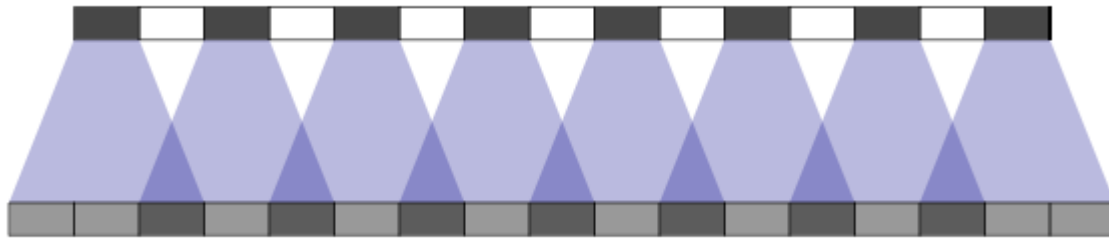
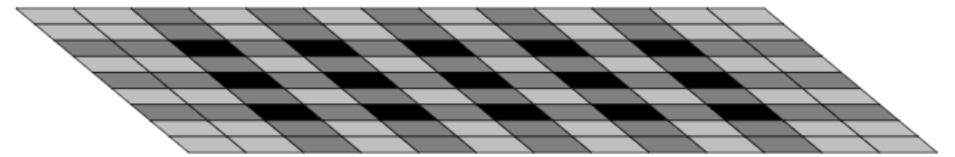
Donahue, et al., 2016 [3]



Dumoulin, et al., 2016 [4]



Unfortunately, deconvolution can easily have “uneven overlap,” putting more of the metaphorical paint in some places than others [7].



AutoEncoders Convolucionales

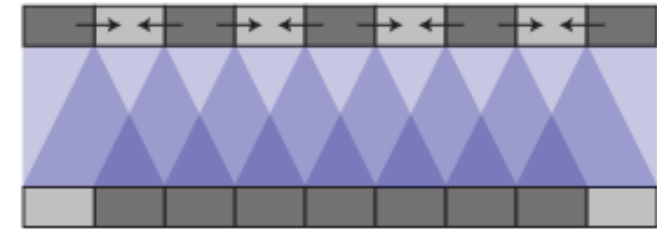


Using deconvolution.
Heavy checkerboard artifacts.



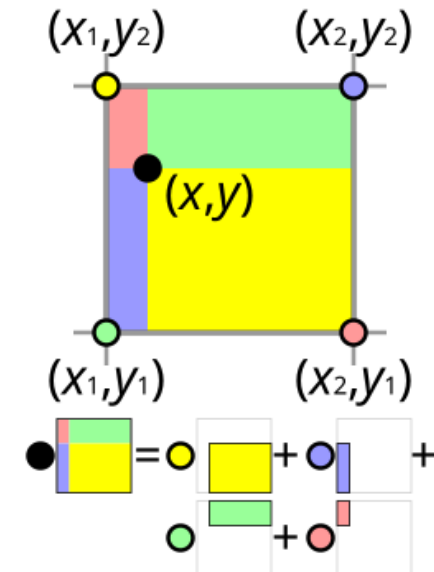
Using resize-convolution.
No checkerboard artifacts.

Resize + con2D



$$\begin{bmatrix} a + \frac{1}{2}b & \frac{1}{2}b + c \\ \frac{1}{2}a & \frac{1}{2}a + b + \frac{1}{2}c & \frac{1}{2}c \\ & a + \frac{1}{2}b & \frac{1}{2}b + c \\ & \frac{1}{2}a & \frac{1}{2}a + b + \frac{1}{2}c & \frac{1}{2}c \end{bmatrix}$$

Bilinear-Resize Convolution



AutoEncoders Reducción de dimensionalidad

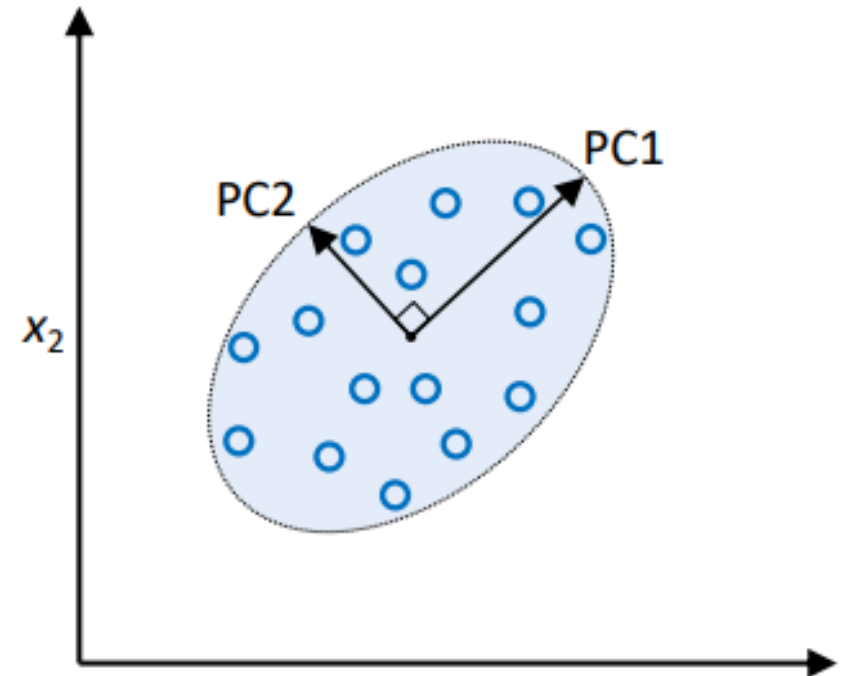
Aprendizaje no supervisado

Trabajar con conjuntos de datos sin considerar la variable de destino

Algunas aplicaciones y objetivos:

- Encontrar estructuras ocultas en los datos
- Compresión de datos
- Agrupación
- Recuperar objetos similares
- Análisis exploratorio de datos
- Generando nuevos ejemplos

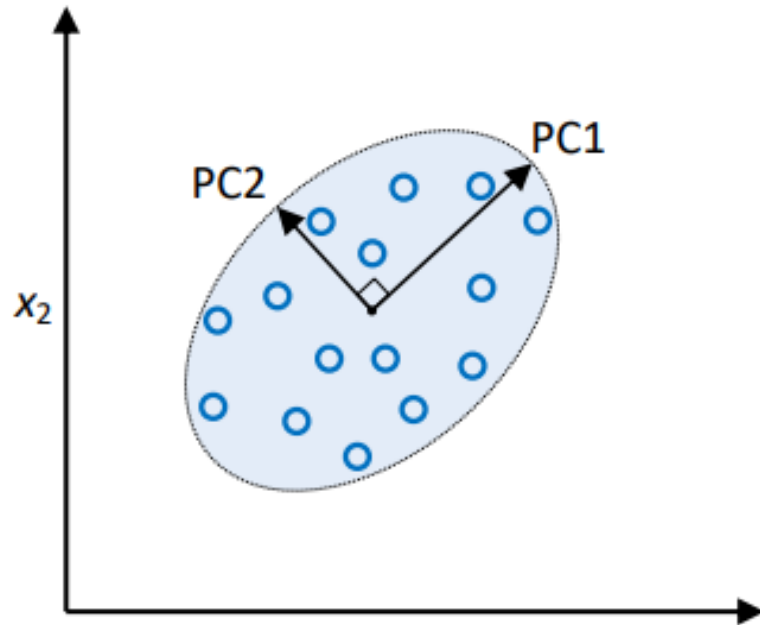
Análisis de componentes principales (PCA por sus siglas en inglés)



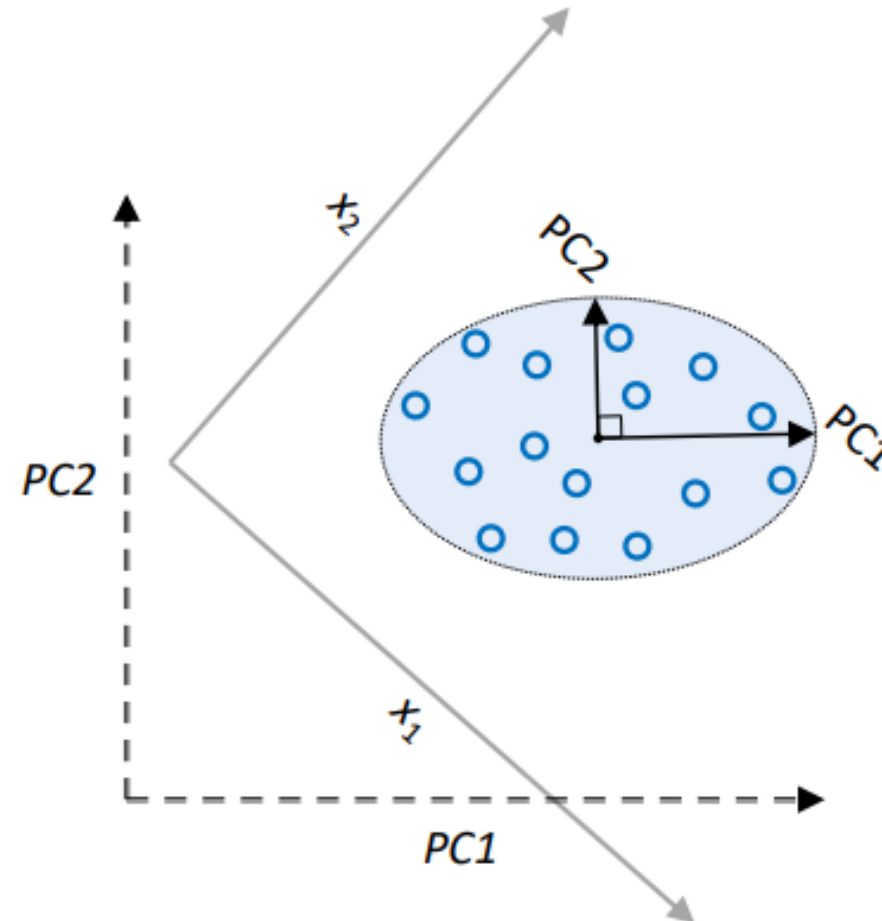
AutoEncoders Reducción de dimensionalidad

Análisis de componentes principales (PCA por sus siglas en inglés)

Encontrar direcciones de mayor varianza

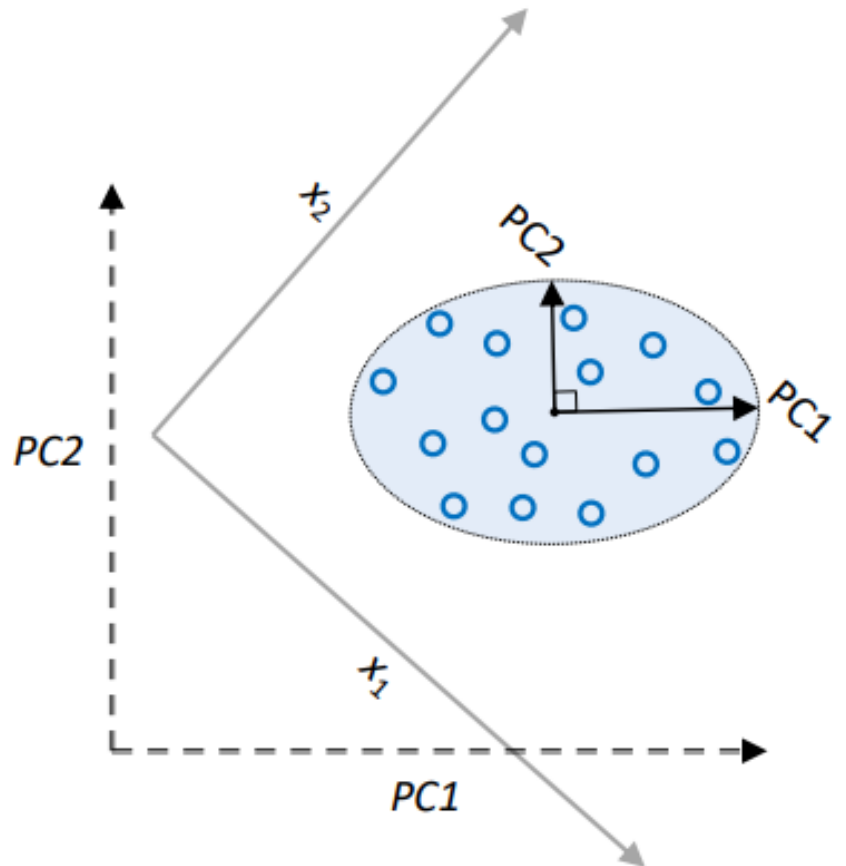


Transformar características en direcciones de máxima variación.



AutoEncoders Reducción de dimensionalidad

Análisis de componentes principales
(PCA por sus siglas en inglés)



Por lo general, considere el subconjunto de
vectores propios de mayor varianza
(reducción de dimensionalidad)

